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Coláiste Náisiúnta na hÉireann | National College of Ireland

This certificate is awarded to:

Lestat Azariel Alvarez Quintana Ordiz

FOR COMPLETING THE PROGRAMME

Academic Integrity: For Students, Second Edition

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Project Report CA1

Module: Data Structures and Algorithms

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Git Hub Repository Link: https://github.com/LestatAQO/Data-Structures-Algorithms-Project-CA1-.git

Proposal

From the options given to us for this project of the Smart Docklands Pilot Call Themes. I choose Urban Greening/Biodiversity.

The cities are starting to expand and grown, but while doing so we are removing most of our green spaces. These green areas are important for the ecosystem and for us, humans. The more trees an area have the better air quality that area receives which would improve our life. The trees are also a natural cover from rain which Ireland suffers from around all year long.

The main problem I want to tackle is that in cities it seems that they have a hard time tracking and managing the new plantations or the state of the trees. Even though watering might not be a huge problem because of the amount of rain, a few dry days can affect the life of a growing tree and it's important to visit them and service them making sure they are not having any mould or something that will affect their life spam.

I work in a hotel; on there we have a back garden that is taken care of regularly by a gardener that comes every few days and waters, when necessary, does plant maintenance checking by checking their state and fixing their problems. This is done by following an on-paper schedule and a map of what and where something is located on the garden.

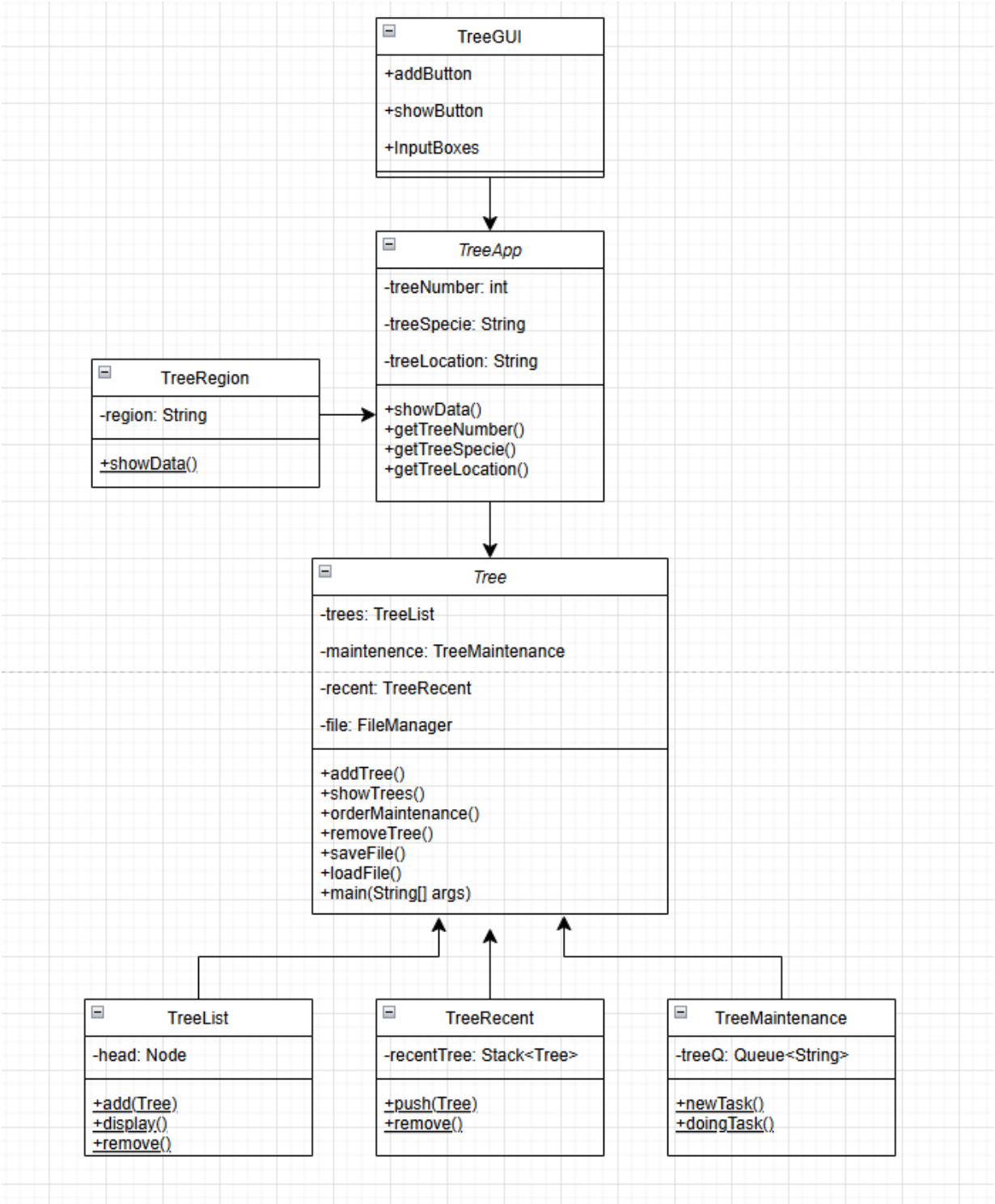
This could be done digitally to some extent to help cities maintain their green spaces healthy.

The plan is to make an application on Java with NetBeans. That will allow users to record the new trees that are planted and store data related to them like, tree species, location and date of planting. While also allowing them to plan when to check them, see all trees and modify/remove data.

I will be using a single Linked List to store the trees data, a stack to store the recently planted and removing the most recent entries. And a queue that will handle tasks using FIFO for the maintenance of the trees.

With this simple tool I aim to facilitate the management of these green areas and helping the environment.

Class Diagram:


TreeGUI
+addButton
+showButton
+InputBoxes
TreeApp
-treeNumber: int
-treeSpecie: String
-treeLocation: String
+showData()
+getTreeNumber()
+getTreeSpecie()
+getTreeLocation()
TreeRegion
-region: String
+showData()
Tree
-trees: TreeList
-maintenence: TreeMaintenance
-recent: TreeRecent
-file: FileManager
+addTree()
+showTrees()
+orderMaintenance()
+removeTree()
+saveFile()
+loadFile()
+main(String[] args)
TreeList
-head: Node
+add(Tree)
+display()
+remove()
TreeRecent
-recentTree: Stack<Tree>
+push(Tree)
+remove()
TreeMaintenance
-treeQ: Queue<String>
+newTask()
+doingTask()